Seoha Kim

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Research Interests

My research aims to enable embodied AI to perceive and reason about dynamic 3D environments. My interests include 4D human–object interaction understanding, functional 3D scene understanding, world models, and vision–language–action (VLA) models.

Education

University of Melbourne — Feb 2026 – Present
Graduate Researcher, Computing and Information Systems — Melbourne, Australia
Advisors: Prof. Taehyun Rhee, Prof. Kris Ehinger

Yonsei University — Mar 2022 – Aug 2024
Master, Artificial Intelligence — Seoul, South Korea
Advisor: Prof. Youngjung Uh

Yonsei University — Mar 2015 – Feb 2021
Bachelor, double major in Business Administration and Cognitive Science — Seoul, South Korea

Selected Publications

Hand-4DGS: Feed-Forward 3D Gaussian Splatting for 4D Hand Reconstruction from Egocentric Videos — Preprint
Jeongmin Bae*, **Seoha Kim***, Marc Pollefeys, Mahdi Rad, Youngjung Uh, Taein Kwon

Per-Gaussian Embedding based Deformation for Deformable 3D Gaussian Splatting — ECCV 2024
Jeongmin Bae*, **Seoha Kim***, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh

Optimizing Dynamic NeRF and 3DGS with No Video Synchronization — ECCVW 2024 Wild3D
Seoha Kim*, Jeongmin Bae*, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh

Sync-NeRF: Generalizing Dynamic NeRFs to Unsynchronized Videos — AAAI 2024
Seoha Kim*, Jeongmin Bae*, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh

Co-Authored Publications

Efficient 4D Scaffold Gaussian Splatting with Dynamic-Aware Anchor Growing — AAAI 2026
Woong Oh Cho, In Cho, **Seoha Kim**, Jeongmin Bae, Youngjung Uh, Seon Joo Kim

Compensating Spatiotemporally Inconsistent Observations for Online Dynamic 3D Gaussian Splatting — SIGGRAPH 2025
Youngsik Yun, Jeongmin Bae, Hyunseung Son, **Seoha Kim**, Hahyun Lee, Gun Bang, Youngjung Uh

Rethinking Open-Vocabulary Segmentation of Radiance Fields in 3D Space — AAAI 2025
Hyunjee Lee*, Youngsik Yun*, Jeongmin Bae, **Seoha Kim**, Youngjung Uh

* Equal contribution.

Research Experience

Research Collaboration Sep 2024 – Aug 2026
With researchers from Meta Reality Labs and Microsoft Zurich Remote

- Proposed feed-forward Gaussian Splatting for dynamic 3D hand reconstruction from egocentric videos.

Research Collaboration Jan 2023 – Jun 2024
ETRI (National Research Institute) Seoul, South Korea

- Proposed dynamic Gaussian Splatting representations for high-quality view synthesis in dynamic scenes.
- Developed a method for training dynamic NeRFs on unsynchronized multi-view videos.

Research Collaboration Feb 2022 – Dec 2022
LG Display Seoul, South Korea

- Developed a knowledge distillation method for panel defect classification using patch-level feature similarity.

Work Experience

Research Fellow Feb 2025 – Dec 2025
ETRI (National Research Institute) Seoul, South Korea

- Improved dynamic 3D hand and pose accuracy using multiple egocentric videos.

Machine Learning Engineer Mar 2021 – Aug 2021
Plask (AI Startup)

- Improved video-based 3D pose estimation by reviewing and implementing state-of-the-art methods.

Data Scientist Intern Sep 2019 – Feb 2020
Hyundai Mobis

- Improved defect classification on factory datasets including motor noise, CT patterns, and solder lines.

Academic Service

Peer Reviewer: CVPR, ECCV, NeurIPS, AAAI, TPAMI, reviewing submissions in 3D reconstruction and generative model

Patents

Method and Apparatus for Representing Dynamic Neural Radiance Fields from Unsynchronized Videos 2026
U.S. Patent, US 12,737,967 B2 (granted Sep 15, 2026)

Method and Apparatus for Decoding Image 2025
U.S. Patent Application, US 2025/0308078 A1

Selected Honors

1st Place, Minister's Award, AID Korea 2021
Animal Datathon Korea – Cow keypoint detection

Silver Medal (Top 2%), Kaggle 2021
Cassava leaf disease classification